# 05 · THE THREE PILLARS OF LOSS FUNCTIONS

## Mathematical Formulations, Backpropagation Dynamics, and Gradient Roles

**Project:** SAR-Guided Satellite Cloud Removal · ECRformer + Conditional PatchGAN
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**Document Purpose & The Tripartite Loss Architecture**

In satellite optical reconstruction, no single loss function can solve cloud removal alone. A pure pixel loss produces blurry smudges; a pure structural loss distorts spectral reflectance; a pure adversarial loss hallucinates non-existent buildings. Our pipeline balances **Three Fundamental Pillars**:

1. **Pillar 1: Photometric / Radiometric Bedrock** ($\mathcal{L}_{L1}, \mathcal{L}_{\text{projs}}$) $\implies$ Preserves multi-spectral color and photon energy.
2. **Pillar 2: Structural & Frequency Scaffold** ($\mathcal{L}_{\text{SDFL}}, \mathcal{L}_{\text{grad}}, \mathcal{L}_{\text{corr}}$) $\implies$ Recovers sharp physical edges and SAR-aligned contours.
3. **Pillar 3: Adversarial & Perceptual Finisher** ($\mathcal{L}_{\text{adv}}, \mathcal{L}_{\text{FM}}$) $\implies$ Synthesizes natural high-frequency micro-texture.

## 1 Master Loss Formulation & Component Matrix

The complete generator optimization objective combines all three pillars under soft cloud mask weighting:

$$\mathcal{L}_G = \mathcal{L}_{L1} + \lambda_{\text{projs}}\mathcal{L}_{\text{projs}} + \lambda_{\text{SDFL}}\mathcal{L}_{\text{SDFL}} + \lambda_{\text{grad}}\mathcal{L}_{\text{grad}} + \lambda_{\text{corr}}\mathcal{L}_{\text{corr}} + w_{\text{adv}}\mathcal{L}_{\text{adv}} + w_{\text{FM}}\mathcal{L}_{\text{FM}} \quad (1)$$

| Pillar | Loss Component | Weight ($\lambda$) | Mathematical Focus | Primary Failure Prevented |
|---|---|---|---|---|
| **1. Photometric** | Masked $L_1$ ($\mathcal{L}_{L1}$) | 1.00 | Low-frequency absolute reflectance | Spectral color drift & energy bias |
| **1. Photometric** | Multi-Scale Projections ($\mathcal{L}_{\text{projs}}$) | 0.25 | Early decoder stage alignment | Vanishing gradients in early transformer layers |
| **2. Structural** | Scale-Decoupled Frequency ($\mathcal{L}_{\text{SDFL}}$) | 0.50 | Multiscale Laplacian frequency octaves | Blurry conditional mean regression |
| **2. Structural** | Sobel Gradient Magnitude ($\mathcal{L}_{\text{grad}}$) | 0.20 | 1st-order spatial derivatives | Washed-out road and canal boundaries |
| **2. Structural** | Edge Pearson Correlation ($\mathcal{L}_{\text{corr}}$) | 0.15 | Directional gradient alignment | Misaligned geometric features |
| **3. Adversarial** | PatchGAN LSGAN ($\mathcal{L}_{\text{adv}}$) | 0.04 | $16 \times 16$ local patch realism | Flat synthetic grey appearance |
| **3. Adversarial** | Feature Matching ($\mathcal{L}_{\text{FM}}$) | 0.35 | Multi-layer critic activations | GAN checkerboard noise & mode collapse |

Table 1: Comprehensive breakdown of the seven active loss components across the Three Pillars.

# 2 Pillar 1: Photometric / Radiometric Reconstruction

## 2.1 1. Mask-Weighted Scale-Decoupled Spatial $L_1$ Loss

$$\mathcal{L}_{L1} = \frac{1}{\|M\|_1 \cdot C} \sum_{c=1}^{C} \sum_{i,j} M(i,j) \cdot |P_{\text{dir}}(c,i,j) - Y(c,i,j)| \tag{2}$$

- **Mathematical Role:** Directly minimizes absolute photon reflectance deviation across all four multi-spectral channels (Red, Green, Blue, NIR).
- **Backpropagation Gradient:**

$$\frac{\partial \mathcal{L}_{L1}}{\partial P_{\text{dir}}(c,i,j)} = \frac{M(i,j)}{\|M\|_1 \cdot C} \cdot \text{sign}\left(P_{\text{dir}}(c,i,j) - Y(c,i,j)\right) \tag{3}$$

  The constant $\pm 1$ sign gradient ensures stable, non-exploding updates even when large cloud illumination spikes occur.

## 2.2 2. Auxiliary Multi-Scale Projection Loss ($\mathcal{L}_{\text{projs}}$)

Intermediate representations from Decoder Stages 1 and 2 ($32 \times 32$ and $64 \times 64$) are projected to 4 channels via $1 \times 1$ convolutions ($P^{(s)}$) and supervised against downsampled ground truth $Y^{(s)}$:

$$\mathcal{L}_{\text{projs}} = \sum_{s \in \{1,2\}} \frac{1}{2^s} \left\| M^{(s)} \odot \left( P^{(s)} - Y^{(s)} \right) \right\|_1 \tag{4}$$

**Gradient Role:** Injects deep supervision directly into early transformer layers, bypassing vanishing gradients.

# 3 Pillar 2: Structural & Frequency Edge Restoration

## 3.1 1. Scale-Decoupled Frequency Loss (SDFL)

SDFL constructs an $S$-level Laplacian pyramid of both prediction $P_{\text{dir}}$ and ground truth $Y$. The high-frequency band at octave $s$ is $H^{(s)} = I^{(s)} - \text{Upsample}(\text{Downsample}(I^{(s)}))$:

$$\mathcal{L}_{\text{SDFL}} = \sum_{s=1}^{S} 2^{S-s} \cdot \left\| \nabla H_P^{(s)} - \nabla H_Y^{(s)} \right\|_1 \tag{5}$$

- **Mathematical Role:** Weights fine spatial octaves with exponentially higher importance ($2^{S-s}$).
- **Gradient Role:** Specifically punishes the loss of high-frequency Fourier components, forcing the generator to inpaint fine roads and field borders rather than flat averages.

## 3.2 2. Directional Sobel Gradient Loss ($\mathcal{L}_{\text{grad}}$)

Using 2D horizontal ($K_x$) and vertical ($K_y$) Sobel convolution kernels:

$$\nabla_x I = K_x * I, \qquad \nabla_y I = K_y * I, \qquad \|\nabla I\| = \sqrt{(\nabla_x I)^2 + (\nabla_y I)^2 + \epsilon} \tag{6}$$

$$\mathcal{L}_{\text{grad}} = \frac{1}{\|M\|_1} \sum_{i,j} M(i,j) \cdot \left| \|\nabla P_{\text{dir}}(i,j)\| - \|\nabla Y(i,j)\| \right| \tag{7}$$

### 3.3 3. Masked Edge Pearson Correlation Loss ($\mathcal{L}_{\text{corr}}$)

Measures the cosine similarity of spatial gradient vectors under the cloud mask:

$$r(\nabla P, \nabla Y) = \frac{\sum(\|\nabla P\| - \mu_P)(\|\nabla Y\| - \mu_Y)}{\sqrt{\sum(\|\nabla P\| - \mu_P)^2}\sqrt{\sum(\|\nabla Y\| - \mu_Y)^2}}, \qquad \mathcal{L}_{\text{corr}} = 1 - r(\nabla P, \nabla Y) \tag{8}$$

Forces predicted edges to be **collinear and co-located** with the true physical ground boundaries.

# 4 Pillar 3: Conditional Adversarial & Perceptual Critique

## 4.1 1. Least-Squares PatchGAN Adversarial Loss ($\mathcal{L}_{\text{adv}}$)

Evaluated over the $8 \times 8$ grid of local $16 \times 16$ spatial patches:

$$\mathcal{L}_D = \frac{1}{2}\mathbb{E}\left[(D(X_D, Y) - 1)^2\right] + \frac{1}{2}\mathbb{E}\left[(D(X_D, P_{\text{dir}}))^2\right] \qquad \text{(Critic Step)} \tag{9}$$

$$\mathcal{L}_{\text{adv}}(G) = \mathbb{E}\left[(D(X_D, P_{\text{dir}}) - 1)^2\right] \qquad \text{(Generator Step)} \tag{10}$$

**Gradient Role:** Injects high-frequency stochastic textures (grassland turbulence, roof tile patterns) where deterministic losses cannot guess exact pixel values.

## 4.2 2. Multi-Layer Feature Matching Loss ($\mathcal{L}_{\text{FM}}$)

Instead of relying on ImageNet pre-trained VGG networks (which are completely blind to satellite infrared and SAR radar channels), we extract internal feature maps directly from the $L = 4$ intermediate convolutional layers of our trained Discriminator $D_\phi$:

$$\mathcal{L}_{\text{FM}}(G, D) = \sum_{l=1}^{L} \frac{1}{N_l} \left\| D^{(l)}(X_D, Y) - D^{(l)}(X_D, P_{\text{dir}}) \right\|_1 \tag{11}$$

**Gradient Role:** Acts as an adaptive, multi-modal perceptual loss that stabilizes the adversarial game and completely eliminates high-frequency checkerboard artifacts.

# 5 Why We Used These Methods Over Others

### 1. Why $L_1$ (MAE) over $L_2$ (MSE) for Pillar 1?

**Alternative:** Mean Squared Error $\mathcal{L}_{L2} = (P - Y)^2$.

- **Failure of Alternative:** $L_2$ penalizes large errors quadratically ($(\Delta)^2$). Under dense clouds where ground uncertainty is high, $L_2$ hedges by predicting the conditional mean $\mathbb{E}[Y|X]$, guaranteeing a blurry grey output. Furthermore, outliers create massive gradient spikes ($2\Delta$), causing training divergence.
- **Why Our Choice Wins:** $L_1$ penalizes errors linearly, which corresponds mathematically to finding the **conditional median**. This preserves sharp transitions across boundaries and provides bounded constant gradients ($\pm 1$).

### 2. Why Scale-Decoupled Frequency Loss (SDFL) over Standard SSIM Alone for Pillar 2?

**Alternative:** Structural Similarity Index Measure (SSIM) alone ($\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(P, Y)$).

- **Failure of Alternative:** SSIM is computed inside small local Gaussian windows ($11 \times 11$). When an entire $100 \times 100$ region is opaque cloud, local SSIM saturates to zero and provides almost flat gradients, failing to enforce edge sharpness.
- **Why Our Choice Wins:** SDFL operates across multiple dyadic scales via Laplacian pyramids. It isolates high frequencies into dedicated octave bands and multiplies their gradients by $2^{S-s}$, guaranteeing that fine road and water contours receive dominant backpropagation force.

**3. Why Discriminator Feature Matching over Pre-Trained VGG Perceptual Loss for Pillar 3?**

**Alternative:** VGG-16 / VGG-19 perceptual loss pre-trained on ImageNet.

- **Failure of Alternative:** ImageNet VGG is trained exclusively on 3-channel optical RGB photographs of dogs, cars, and consumer objects. It has **zero understanding** of Near-Infrared (Band 8) or Sentinel-1 SAR microwave radar backscatter. Forcing multi-spectral satellite data through RGB VGG causes severe spectral distortion.
- **Why Our Choice Wins:** Our PatchGAN critic natively accepts the complete 11-channel multi-modal tensor $[S_1, S_2, M, \text{Candidate}]$. Its internal activations understand radar backscatter and infrared absorption natively, providing scientifically grounded perceptual supervision.

## 6 15-Second Viva Examiner Defense Quick-Sheet

**Q: What prevents your GAN from hallucinating fake terrain features?**
*Answer:* The 11-channel conditional input of the Discriminator and the high weighting of Pillar 1 ($L_1 = 1.0$) and Pillar 2 (SDFL = 0.50). The adversarial loss ($w_{\text{adv}} = 0.04$) only controls fine texture; large-scale geometry is anchored to the physical SAR radar signal.

**Q: Why are your losses weighted by the cloud mask $M$?**
*Answer:* Mask-weighting concentrates gradient updates exclusively on pixels that actually require reconstruction. Clear-sky pixels already have perfect ground truth in $S_{2,\text{cloudy}}$ and are preserved via alpha blending.